

COHERE™ PLASTOMER S100

METALLOCENE POLYOLEFIN PLASTOMER

DESCRIPTION

COHERE™ Metallocene Polyolefin Plastomer (POP) S100 is an ethylene-octene copolymers produced via solution polymerization using advanced metallocene catalyst. It performs well in high performance flexible packaging applications with excellent toughness, hot tack, hot seal, optical properties and enhanced packaging integrity to reduce loss/waste.

TYPICAL APPLICATIONS

- Packaging with outstanding sealing performance (low seal initiation temperature, broad hot tack window, excellent seal through contamination).
- Packaging available for high speed form-fill-sealing machine.

TYPICAL PROPERTY VALUES

Revision 20230403

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
POLYMER PROPERTIES			
Density	900	kg/m ³	ASTM D792
Melt Flow Rate (MFR)			
at 190°C and 2.16 kg	1.0	g/10 min	ASTM D1238
OPTICAL PROPERTIES			
Haze	2.7	%	ASTM D1003
FILM PROPERTIES			
Tensile Properties ⁽¹⁾			
stress at break, MD	50	MPa	ASTM D882
stress at break, TD	55	MPa	ASTM D882
strain at break, MD	620	%	ASTM D882
strain at break, TD	670	%	ASTM D882
1% secant modulus, MD	50	MPa	ASTM D882
1% secant modulus, TD	56	MPa	ASTM D882
Dart Impact Strength ⁽¹⁾			
Method A	>1000	g	ASTM D1709
Elmendorf Tear Strength ⁽¹⁾			
MD	10	g/μm	ASTM D1922
TD	17	g/μm	ASTM D1922
Sealing Initiation Temperature ⁽²⁾	78	°C	SABIC method
THERMAL PROPERTIES			
Melting Point	107	°C	SABIC method

(1) Properties have been measured by producing 40 μm film with 2.1 BUR, screw diameter 35mm, die diameter 100mm, die gap 1mm.

(2) Temperature at which 4 N/25.4mm heat seal strength is achieved.

PROCESSING CONDITIONS

Typical processing conditions for COHERE™ S100 are:

Barrel temperature: 180 - 200°C

Blow up ratio: 2.0 - 3.0

© 2023 Copyright by SABIC. All rights reserved

CHEMISTRY THAT MATTERS™



STORAGE AND HANDLING

The resin should be stored in a manner to prevent a direct exposure to sunlight and / or heat. The storage area should also be dry and preferably do not exceed 50°C. SABIC® would not give warranty to bad storage conditions that may lead to quality deterioration such as color change, bad smell and inadequate product performance. It is advisable to process PE resin within 6 months after delivery.

DISCLAIMER

Any sale by SABIC, its subsidiaries and affiliates (each a "seller"), is made exclusively under seller's standard conditions of sale (available upon request) unless agreed otherwise in writing and signed on behalf of the seller. While the information contained herein is given in good faith, SELLER MAKES NO WARRANTY, EXPRESS OR IMPLIED, INCLUDING MERCHANTABILITY AND NONINFRINGEMENT OF INTELLECTUAL PROPERTY, NOR ASSUMES ANY LIABILITY, DIRECT OR INDIRECT, WITH RESPECT TO THE PERFORMANCE, SUITABILITY OR FITNESS FOR INTENDED USE OR PURPOSE OF THESE PRODUCTS IN ANY APPLICATION. Each customer must determine the suitability of seller materials for the customer's particular use through appropriate testing and analysis. No statement by seller concerning a possible use of any product, service or design is intended, or should be construed, to grant any license under any patent or other intellectual property right.